

# Data Consumption board on Polly

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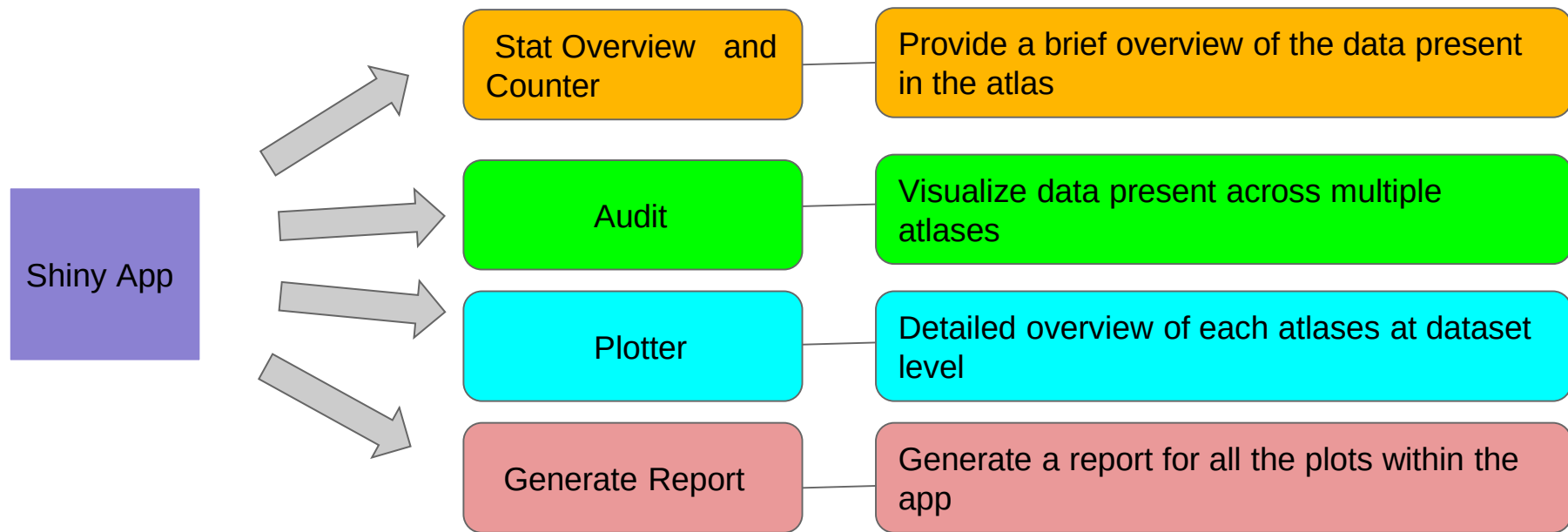


# Agenda

- Overview of modules
- Detailed explanation of each modules:
  1. Stat overview and counter module
  2. Audit module
  3. Plotter module
  4. Generate report module
- Acknowledgement



# Stat Overview and Counter Module



# Stat Overview and Counter module



James



Polly Cross OA

Prerequisite:

- Familiar to SQL

However, James have no prior experience with SQL and hence, cannot fetch data from Polly Cross OA

# Stat Overview and Counter

With **Polly Cross OA**:

API Version

v2

Enter SQL query

```
SELECT COUNT (DISTINCT curated_disease) AS count FROM gdc.datasets"
```

Submit

When you click the button above, you should see the output below update to reflect the value you entered at the top:

[For help, See SQL Query Examples](#)

Elapsed time = --- 31.634050607681274 seconds ---

Query: Successful

Show 10 entries

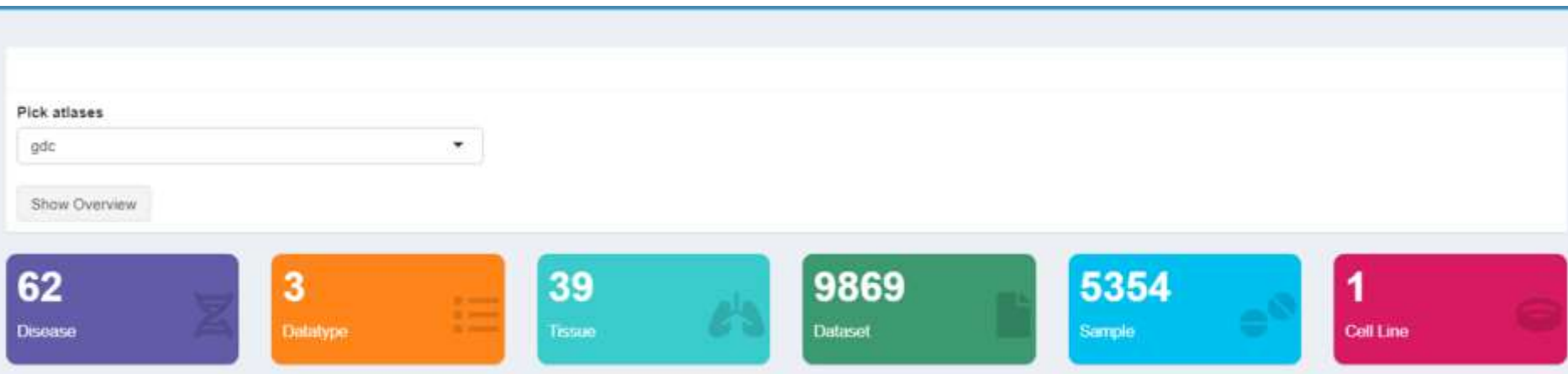
Search:

|   | count |
|---|-------|
| 1 | 63    |

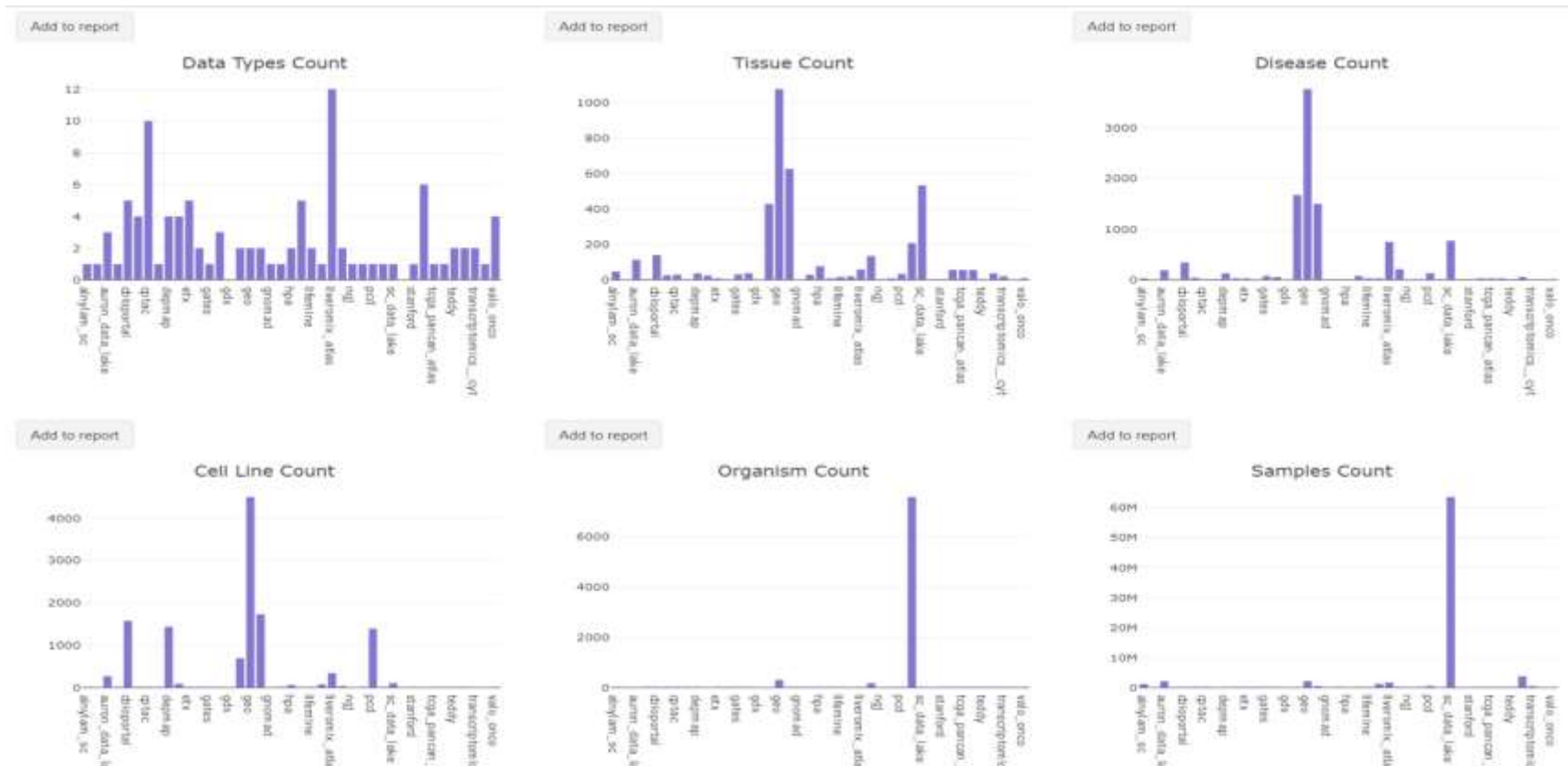


With **Stat overview and counter module**:

1. Overview for a particular atlas



## 2. Overview across multiple atlases



# Audit module



Noah



Polly Cross OA

Prerequisite:

- Familiar to SQL
- Familiar to R/Python for visualization

Let's assume, Noah wants to fetch and visualize the data available for any curated disease/curated tissue/ curated organism across multiple atlases. However, he does not accomplish the desired prerequisite.



# Audit module

Using **Polly Cross OA**:

Enter SQL query

```
SELECT dataset_id, curated_disease, description, data_type, curated_organism, curated_tissue, curated_cell_line,  
       curated_cell_type, curated_drug, total_num_samples, src_repo FROM tcga_datasets WHERE curated_disease LIKE "%Colorectal"
```

Submit

When you click the button above, you should see the output below update to reflect the value you entered at the top:

[For help, See SQL Query Examples](#)

Elapsed time = --- 7.7562432289123535 seconds ---

Query Successful

Show 10 entries

Search:

|   | dataset_id                                | curated_disease        | description                                                     | data_type             | curated_organism | curated_tissue | curated_cell_line | curated_cell_type | curated_drug | total_m |
|---|-------------------------------------------|------------------------|-----------------------------------------------------------------|-----------------------|------------------|----------------|-------------------|-------------------|--------------|---------|
| 1 | COAD_Copy_Number_Segment_TCGA-3L-AA1B-01A | [Colorectal Neoplasms] | Colon adenocarcinoma Copy Number Variation for TCGA-3L-AA1B-01A | Copy Number Variation | ["Homo sapiens"] | [colon]        | [None]            | [None]            | [None]       |         |



## With Audit module:

Pick Atlas

ncgi\_geo

Pick Level

datasets

Pick Column to search query

curated\_disease

Query

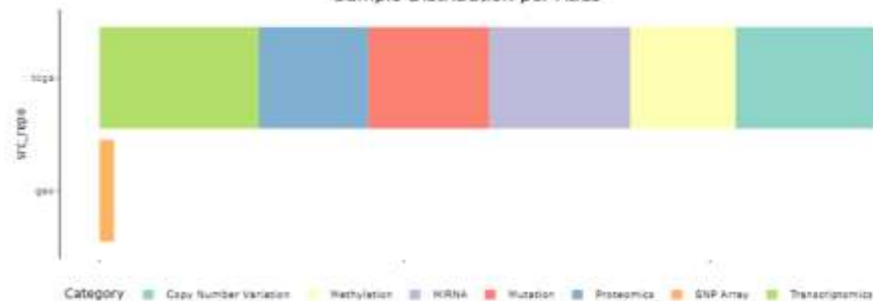
Colorectal Neoplasms

Audit

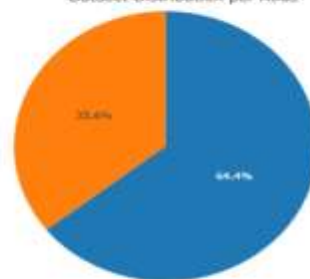
Issue Distribution per Atlas



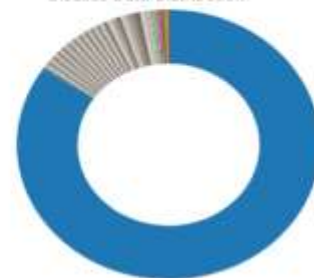
Sample Distribution per Atlas



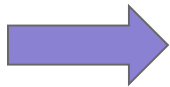
Dataset Distribution per Atlas



Disease Data Distribution



# Plotter module



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Polly Cross OA

Prerequisite:

- Knowledge about Omix Atlas.
- Familiar to SQL
- Familiar to R/Python for visualization

Let's assume, Nancy, a bioinformatician, is looking for gene information for a particular dataset from a specific atlas.

To accomplish this, Nancy can either work with Cross OA platform, or can simply access/visualize data from a single click.



# Plotter Module

With **Polly Cross OA**:

 Polly™ Cross OA Query

API Version

v2

Enter SQL query

SELECT \* FROM data\_matrices.geo\_\_GSE9576\_GPL570

Submit

When you click the button above, you should see the output below update to reflect the value you entered at the top:

[For help, See SQL Query Examples](#)

Elapsed time = --- 4.315646409888403 seconds ---

Query Successful

Show 10 entries

Search:

|   | rid   | gsm242028 | gsm242029 | gsm242030 | gsm242031 | gsm242032 | gsm242033 | gsm242034 | gsm242035 | gsm242036 | gsm242037 | gsm242038 | gsm242039 |
|---|-------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| 1 | RPL30 | 13.7429   | 13.2066   | 13.0164   | 13.2423   | 13.2546   | 13.206    | 13.1952   | 13.1682   | 12.4629   | 13.2975   | 13.2738   | 13.2529   |
| 2 | COPA  | 10.123    | 9.0967    | 8.7624    | 9.3877    | 9.0832    | 8.9727    | 9.0746    | 9.0308    | 9.0441    | 9.4196    | 9.0844    | 9.096     |
| 3 | SCAP  | 7.9088    | 8.0095    | 8.0679    | 7.8841    | 7.8915    | 7.8357    | 7.9792    | 7.8947    | 8.0672    | 7.94      | 8.0372    | 7.9104    |



With the **Plotter** module:

Pick atlases  
geo

Pick search level  
dataset\_id

Enter Dataset Id  
GSE9576\_GPL570

Search Dataset

Show 10 entries

Search:

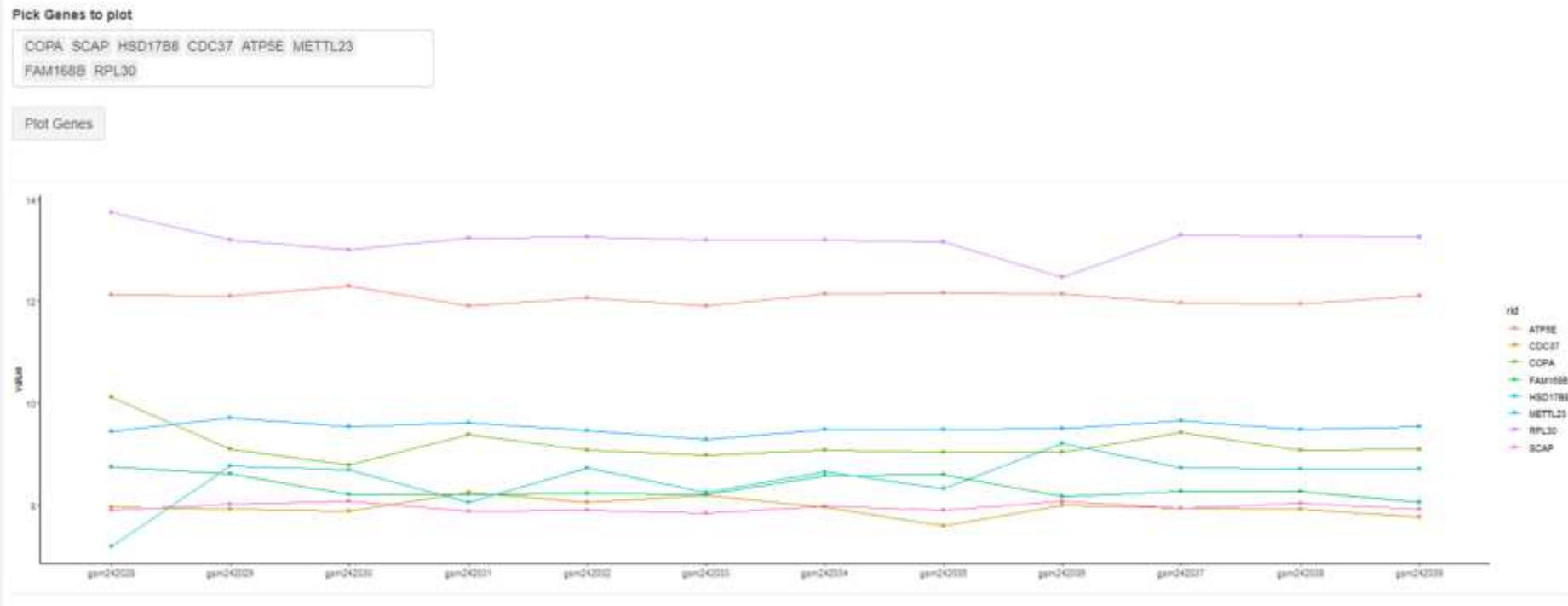
| rid     | gsm242028 | gsm242029 | gsm242030 | gsm242031 | gsm242032 | gsm242033 | gsm242034 | gsm242035 | gsm242036 | gsm242037 | gsm242038 | gsm242039 | row_id  |
|---------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|---------|
| RPL30   | 13.7429   | 13.2066   | 13.0184   | 13.2423   | 13.2546   | 13.206    | 13.1962   | 13.1682   | 12.4629   | 13.2975   | 13.2736   | 13.2529   | rpl30   |
| COPA    | 10.123    | 9.0957    | 8.7824    | 9.3877    | 9.0832    | 8.9727    | 9.0746    | 9.0308    | 9.0441    | 9.4196    | 9.0644    | 9.096     | copa    |
| SCAP    | 7.9088    | 8.0095    | 8.0679    | 7.8841    | 7.8915    | 7.8357    | 7.9792    | 7.8947    | 8.0672    | 7.94      | 8.0372    | 7.9104    | scap    |
| HSD17B8 | 7.1824    | 8.759     | 8.6976    | 8.0454    | 8.7313    | 8.2529    | 8.6602    | 8.3256    | 9.2106    | 8.7293    | 8.7094    | 8.7058    | hsd17b8 |
| CDC37   | 7.9527    | 7.9128    | 7.8894    | 8.24      | 8.0573    | 8.1923    | 7.9571    | 7.6003    | 7.9885    | 7.9457    | 7.922     | 7.7641    | cdc37   |
| ATP5E   | 12.1166   | 12.1005   | 12.3052   | 11.9066   | 12.0563   | 11.9176   | 12.1371   | 12.1539   | 12.1466   | 11.9765   | 11.9463   | 12.1007   | atp5e   |
| METTL23 | 9.4422    | 9.7168    | 9.5454    | 9.6128    | 9.4655    | 9.2923    | 9.4805    | 9.4897    | 9.4947    | 9.6632    | 9.4803    | 9.5419    | mettl23 |
| FAM168B | 8.7562    | 8.6102    | 8.203     | 8.2086    | 8.2196    | 8.2085    | 8.5719    | 8.5859    | 8.174     | 8.2587    | 8.2605    | 8.0473    | fam168b |
| PDLIM5  | 9.3989    | 10.2771   | 10.364    | 10.3433   | 10.3545   | 10.2849   | 10.3454   | 10.342    | 10.2904   | 10.6398   | 10.6643   | 10.4726   | pdlim5  |
| NUP88   | 9.0188    | 8.8407    | 8.1862    | 8.7675    | 8.8358    | 8.7831    | 8.7792    | 8.545     | 8.1824    | 8.8356    | 8.6323    | 8.7909    | nup88   |

Showing 1 to 10 of 20,352 entries

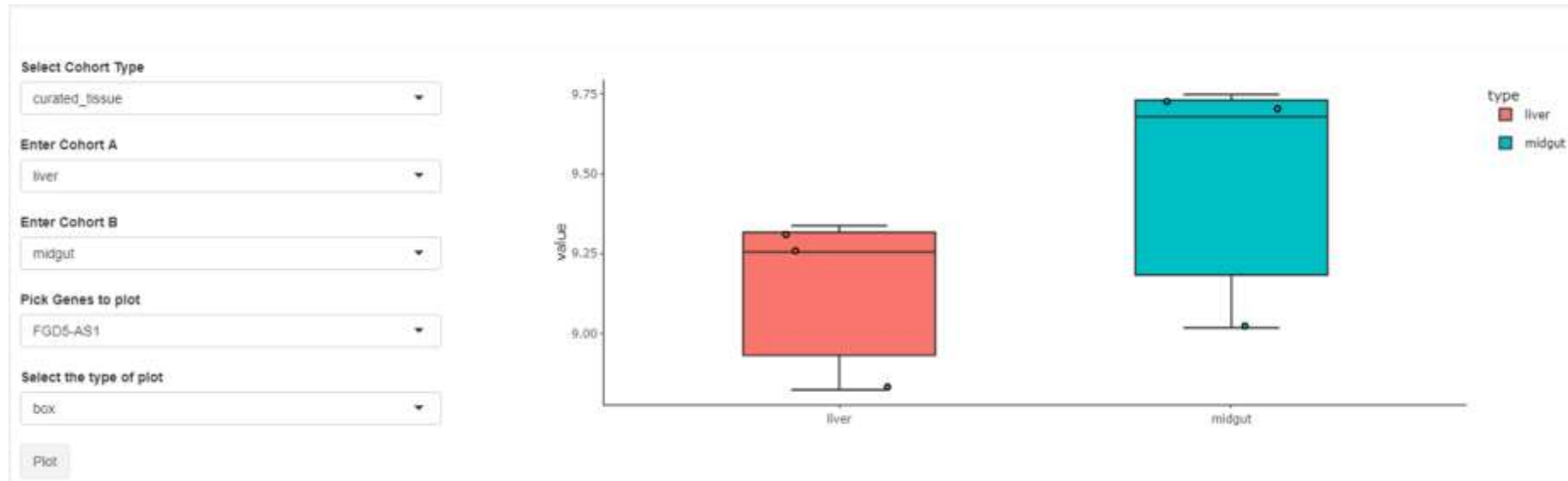
Previous12345...2,036Next



With the **Plotter** module:



With the **Plotter** module:



# Generate Report Module

It's time for some reports after visualizing the data



1 CPLS-DA Data Type Count

2 OPLS-DA Disease Count

3 CPLS-DA Organism Count

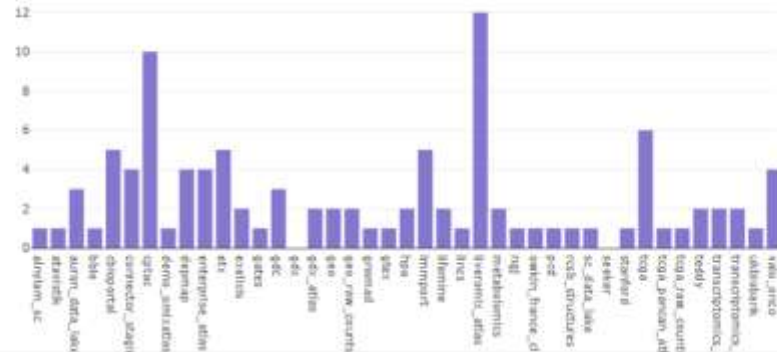
## Journal Club Final Presentation

Elucidate

15 July 2022

## 1 OPLS-DA Data Type Count

### Data Types Count





# Acknowledgement

I would like to express my special thanks of gratitude to the entire BDA team who have helped me to complete this wonderful project, and have helped me to come across various programming skills.

Secondly, I would like to thank the entire Elucidata Family, who have provided me with such a friendly environment.



